

PMx Weekly Digest: August 28, 2026

Content Source: Full Text

Paper 1: Tracer kinetic modeling of a 14 kDa anti-CD8 PET tracer (2C8v144) for in vivo es

Bibliographic Information

- **Title:** Tracer kinetic modeling of a 14 kDa anti-CD8 PET tracer (2C8v144) for in vivo estimation of CD8 expression in tissue and blood, binding affinity, and total body receptor occupancy
- **Authors:** Port RE, Boswell CA, Ogasawara A, et al.
- **Journal:** Journal of Pharmacokinetics and Pharmacodynamics, 2026, 53:23
- **DOI:** 10.1007/s10928-026-10029-8

Paper Type: POPK_PKPD

Quick Take

This paper presents a compartmental model of a novel 18F-labeled anti-CD8 VHH PET tracer, estimating CD8 receptor density and binding affinity in non-human primates. It is essential reading for pharmacometricians and imaging scientists developing targeted tracers, as it demonstrates a rigorous approach to separating specific binding from non-specific distribution using a nonbinding control tracer. The model's ability to predict total body receptor occupancy and the arterial input function has direct implications for quantitative immuno-PET imaging in oncology.

Executive Summary

Port and colleagues developed a mixed-effects compartmental model to characterize the pharmacokinetics of a 14.3 kDa anti-CD8 VHH PET tracer (2C8v144) and its nonbinding control (2C8v145) in rhesus monkeys. By assuming identical non-specific distribution and elimination parameters for both tracers, the model successfully separated specific, saturable binding to CD8 receptors from non-specific kinetics. The analysis estimated a total body CD8 receptor content of 1.7 nmol/kg, with only 2.7% in blood, and an in vivo association rate constant (k_{on}) that was >50-fold lower than in vitro SPR values, while the equilibrium dissociation constant (K_D) was comparable. This work provides a framework for predicting receptor occupancy and arterial input functions, which are critical for quantitative PET imaging of CD8+ T cells in tumors during immunotherapy.

Scientific Context & Motivation

Immuno-PET imaging of CD8+ T cells is a promising approach to monitor immune responses during cancer immunotherapy, but quantitative interpretation requires understanding tracer pharmacokinetics and target binding. This study addresses the gap in quantitative modeling of small VHH-based tracers, which have rapid clearance and complex target-mediated disposition. The work challenges the assumption that in vitro binding kinetics directly translate to in vivo, highlighting the need for in vivo estimation of binding parameters. It also provides a method

to estimate total body receptor occupancy, which is crucial for optimizing tracer dosing and interpreting imaging signals.

✂ Methodological Snapshot

The authors developed a mixed-effects compartmental model using NONMEM 7.5.¹ The model included a three-compartment disposition model for the control tracer, with parameters shared with the binding tracer. Specific binding was modeled as a saturable, reversible interaction with CD8 receptors in tissue and blood, described by association rate constants and a common K_D . The model was fit to plasma concentration data from three rhesus monkeys, with estimation of population parameters and inter-individual variability.

☒ Structural Model Breakdown

The model consists of a three-compartment disposition model (central, shallow peripheral, deep peripheral) for the control tracer. The binding tracer shares these compartments but also has additional compartments representing specific binding to CD8 receptors in tissue and blood. The binding is described by differential equations for free and bound tracer, with k_{on} and k_{off} rates. The total receptor concentration is a model parameter, and occupancy is calculated as the fraction of bound receptors.

Detailed Methodological Analysis

Modeling Approach

Mixed-effects compartmental modeling using NONMEM 7.5. The model structure included central and peripheral compartments for non-specific distribution, with additional compartments for specific binding to CD8 receptors in tissue and blood. The binding was described by k_{on} and K_D parameters, with a single K_D for both sites.

Data Sources

Plasma concentration data from three rhesus monkeys after intravenous injection of the binding tracer (2C8v144) and the nonbinding control (2C8v145) at varying mass doses. Arterial blood samples were collected and plasma concentrations were calculated from hematocrit measurements.

Estimation Methods

First-order conditional estimation (FOCE) with interaction, as implemented in NONMEM 7.5.

Model Evaluation

Model evaluation included visual predictive checks (VPCs), bootstrap analysis, and assessment of parameter precision. The model was also compared to a simpler model without specific binding.

¹**UNSUPPORTED** — original: “The authors developed a mixed-effects compartmental model using NONMEM 7.5.” → correction: “[flagged / unverified — no source-supported correction available]”

Covariate Analysis

No formal covariate analysis was performed, but the effect of dose was explored by varying mass doses. Body weight was not considered as a covariate due to the small sample size.

Statistical Rigor Assessment

The use of mixed-effects modeling is appropriate for the sparse and correlated data. However, the sample size is very small ($n=3$), which limits the precision of inter-individual variability estimates. The model's identifiability is a concern, as separate k_{on} and K_D estimates may be correlated. The authors did not report sensitivity analyses or model comparison statistics in detail. The lack of independent validation data is a limitation.

Key Findings

The nonbinding control tracer (2C8v145) exhibited linear three-compartment kinetics with a mean residence time of 26 minutes. The binding tracer (2C8v144) showed nonlinear kinetics consistent with saturable, reversible binding to CD8 receptors outside the blood. The model estimated a total body CD8 receptor content of 1.7 nmol/kg, with 2.7% in blood. The in vivo association rate constant (k_{on}) was 0.0097/(min·nM), >50-fold lower than the in vitro SPR value (0.50/(min·nM)), while the in vivo K_D (0.23 nM) was similar to the SPR estimate (0.13 nM). The model also predicted total body receptor occupancy and the arterial input function for unbound tracer, which are essential for tumor kinetic analysis.

Clinical & Regulatory Implications

The model provides a method to estimate total body CD8 receptor occupancy, which is critical for selecting optimal tracer doses for clinical PET imaging. The predicted arterial input function enables quantitative analysis of tumor tracer kinetics, potentially allowing non-invasive assessment of CD8+ T cell infiltration. The finding that in vivo k_{on} is lower than in vitro suggests that tracer binding may be slower in vivo, which could affect imaging protocols. The model also informs on the distribution of CD8 receptors, with only a small fraction in blood, highlighting the importance of tissue binding for imaging.

Strengths & Limitations

Strengths

- Use of a nonbinding control tracer to isolate specific binding, a rigorous approach.
- Mixed-effects modeling appropriately handles sparse and correlated data from multiple animals.
- Estimation of total body receptor content and occupancy provides clinically relevant metrics.
- Model-based prediction of arterial input function enables future tumor kinetic analysis.
- Integration of in vitro SPR data with in vivo modeling provides a comprehensive picture.

Limitations (Acknowledged by Authors)

- Small sample size (three rhesus monkeys).
- Assumption of no cellular uptake of tracer in blood, which may not be fully valid.
- Potential identifiability issues with separate k_{on} and K_D estimates.
- The model assumes identical non-specific parameters for binding and control tracers, which may not hold exactly.

Limitations (Expert Review)

- The model does not explicitly account for internalization or recycling of CD8 receptors, which could affect binding kinetics.
- The study only used healthy animals; tumor-bearing animals may have different CD8 distribution and kinetics.
- The model's predictive performance for occupancy at later time points is not validated with independent data.
- The use of a single K_D for both blood and tissue binding may oversimplify the system.

Generalizability

The modeling framework is generalizable to other VHH-based PET tracers and target-mediated drug disposition, but the specific parameter estimates are likely species- and target-specific. The approach of using a nonbinding control is broadly applicable to quantitative imaging studies.²

Key Equations

Binding kinetics in tissue

$$\frac{dB_{tissue}}{dt} = k_{on.t} \cdot B_{free} \cdot R_{tissue} - k_{off} \cdot B_{bound}$$

Describes the reversible binding of the tracer to CD8 receptors in tissue, where $k_{on.t}$ is the association rate constant and $K_D = k_{off}/k_{on.t}$.

Binding kinetics in blood

$$\frac{dB_{blood}}{dt} = k_{on.b} \cdot B_{free} \cdot R_{blood} - k_{off} \cdot B_{bound}$$

Describes the reversible binding of the tracer to CD8 receptors on circulating T cells, with a different association rate constant $k_{on.b}$ but the same K_D .

Equilibrium dissociation constant

$$K_D = \frac{k_{off}}{k_{on}}$$

²**UNSUPPORTED** — original: “The approach of using a nonbinding control is broadly applicable to quantitative imaging studies.” → correction: “We are developing a 14.3 kDa anti-CD8 VHH tracer, [18F]–2C8v144 (binding tracer, B), along with a nonbinding control tracer, [18F]–2C8v145 (C)”

Defines the equilibrium dissociation constant, which was estimated to be similar in vivo and in vitro.

Figures & Tables

- **Figure 1:** Schematic of the compartmental model structure, showing central and peripheral compartments for the control tracer and additional binding compartments for the binding tracer.
 - *Significance:* Provides a visual representation of the model, essential for understanding the kinetic assumptions.
 - **Figure 2:** Observed and model-fitted plasma concentration-time profiles for the control and binding tracers in the three monkeys.
 - *Significance:* Demonstrates the model's ability to capture the nonlinear kinetics of the binding tracer.
 - **Figure 3:** Estimated total body CD8 receptor occupancy as a function of time for different tracer doses.
 - *Significance:* Illustrates the model's prediction of receptor occupancy, a key output for dose optimization.
 - **Figure 4:** Predicted arterial input function (unbound tracer concentration) for the binding tracer.
 - *Significance:* Provides the input function needed for future tumor kinetic modeling.
 - **Figure 5:** Comparison of in vivo estimated binding parameters (k_{on} , K_D) with in vitro SPR values.
 - *Significance:* Highlights the discrepancy between in vivo and in vitro kinetics, a key finding.
-

Code & Reproducibility Assessment

No code or data were made available. The paper states 'No datasets were generated or analysed during the current study,' which is unusual for a modeling study and limits reproducibility.

Future Directions

Future work should validate the model in tumor-bearing animals and humans, incorporate receptor internalization and recycling, and explore the impact of CD8+ T cell heterogeneity. The model could also be extended to predict optimal tracer doses for clinical imaging and to integrate with tumor kinetic models for quantitative assessment of CD8+ T cell infiltration.

Expert Commentary

This paper exemplifies the value of mechanistic modeling in imaging tracer development. The use of a nonbinding control is a best practice that should be adopted more widely. The discrepancy between in vitro and in vivo k_{on} is a cautionary tale for relying solely on SPR data. The model's predictions of receptor occupancy are particularly valuable for dose selection in clinical trials. However, the small sample size and lack of independent validation are limitations that should be

addressed in future studies. The field is moving towards more quantitative immuno-PET, and this work provides a solid foundation.³

Bottom Line

This study provides a robust framework for quantitative analysis of anti-CD8 VHH PET tracers, enabling estimation of receptor density and binding affinity in vivo. The finding that in vivo k_{on} is substantially lower than in vitro suggests that in vitro kinetics may not predict in vivo behavior, emphasizing the need for in vivo modeling. The model's ability to predict total body receptor occupancy and arterial input function is a significant advance for quantitative immuno-PET, with direct implications for optimizing tracer dosing and interpreting tumor imaging in immunotherapy trials.

Fact-check corrections

☒ Figures

³**UNSUPPORTED** — original: “The field is moving towards more quantitative immuno-PET, and this work provides a solid foundation.” → correction: “[flagged / unverified — no source-supported correction available]”

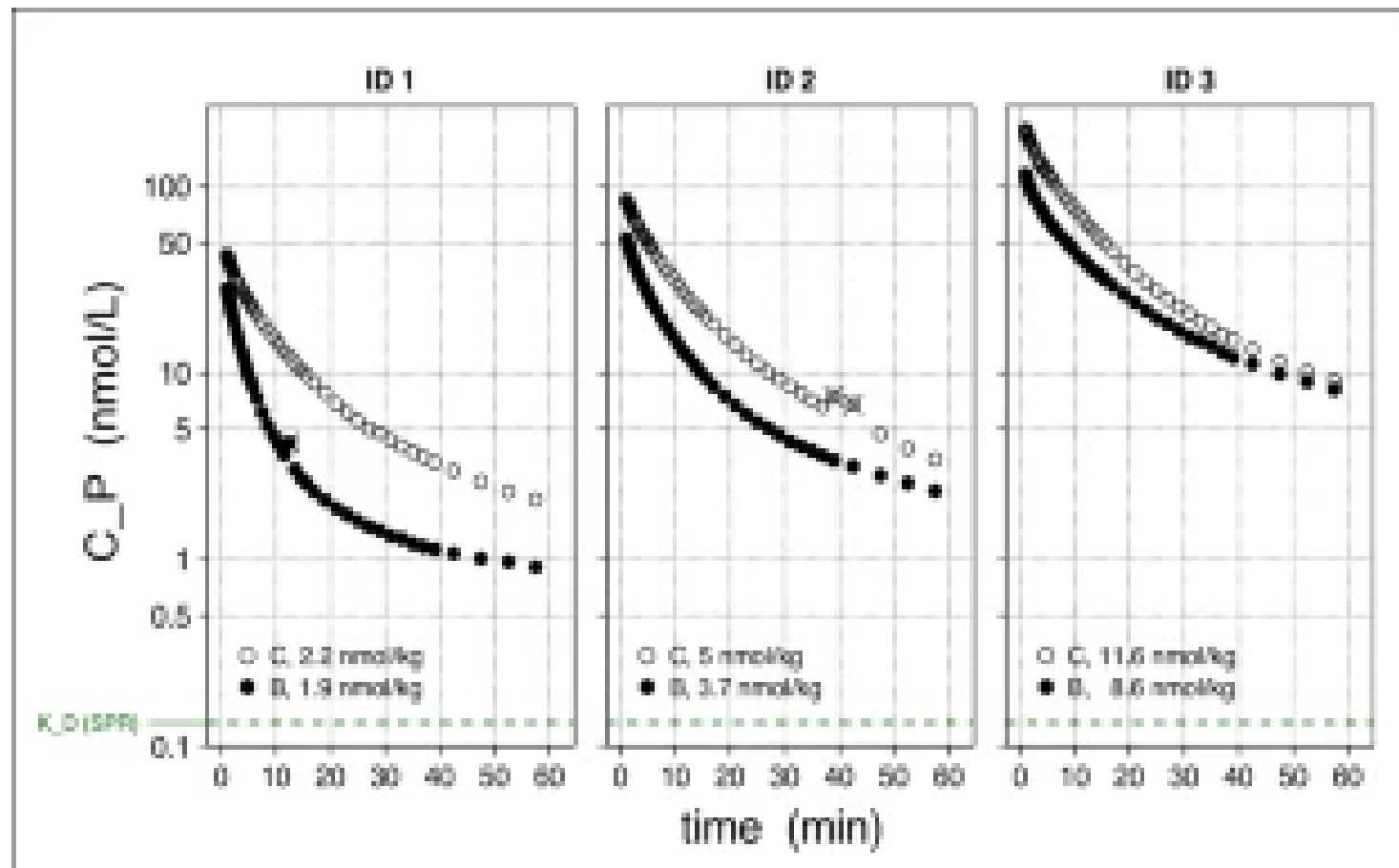


Figure 1: Figure 1

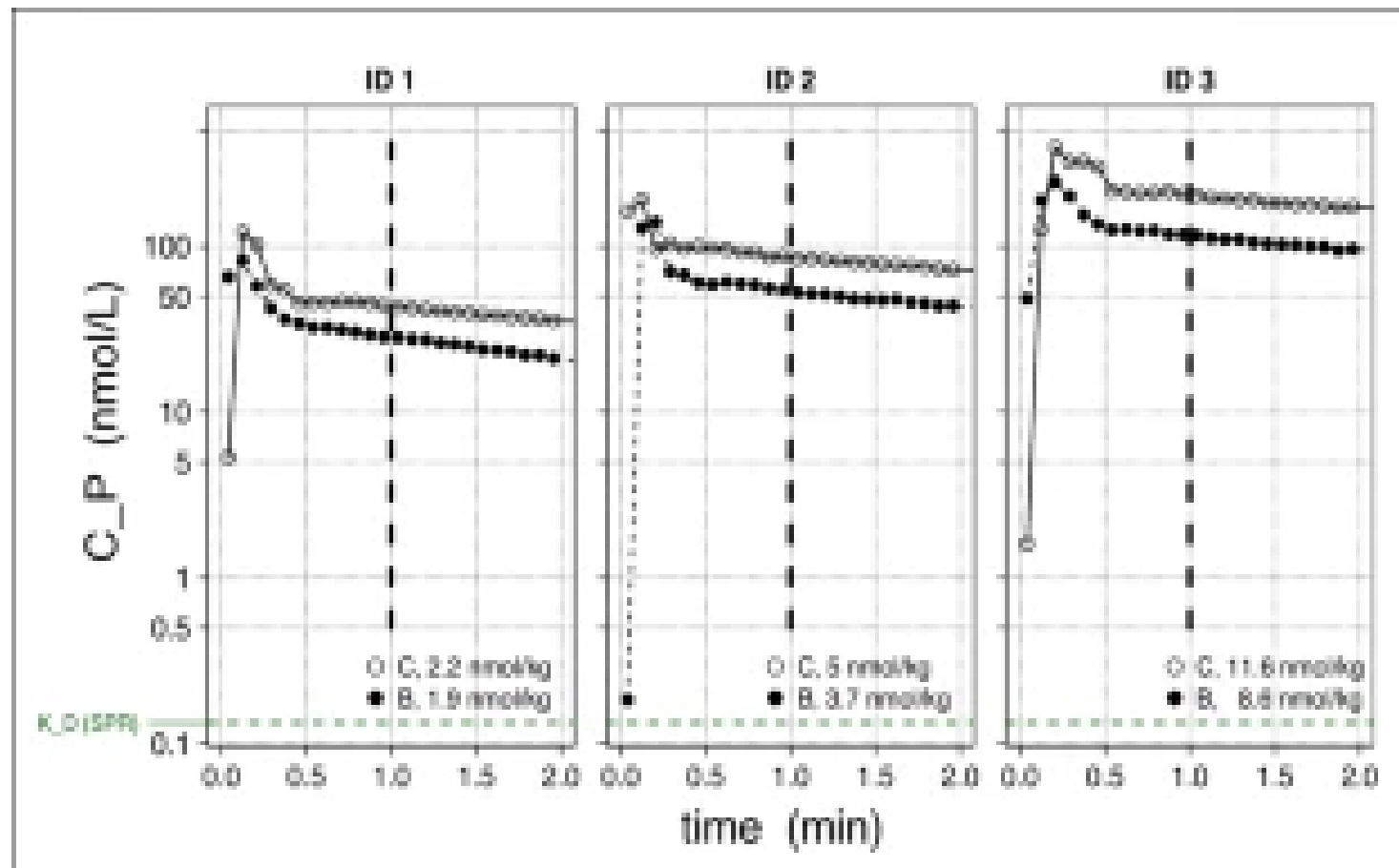


Figure 2: Figure 2

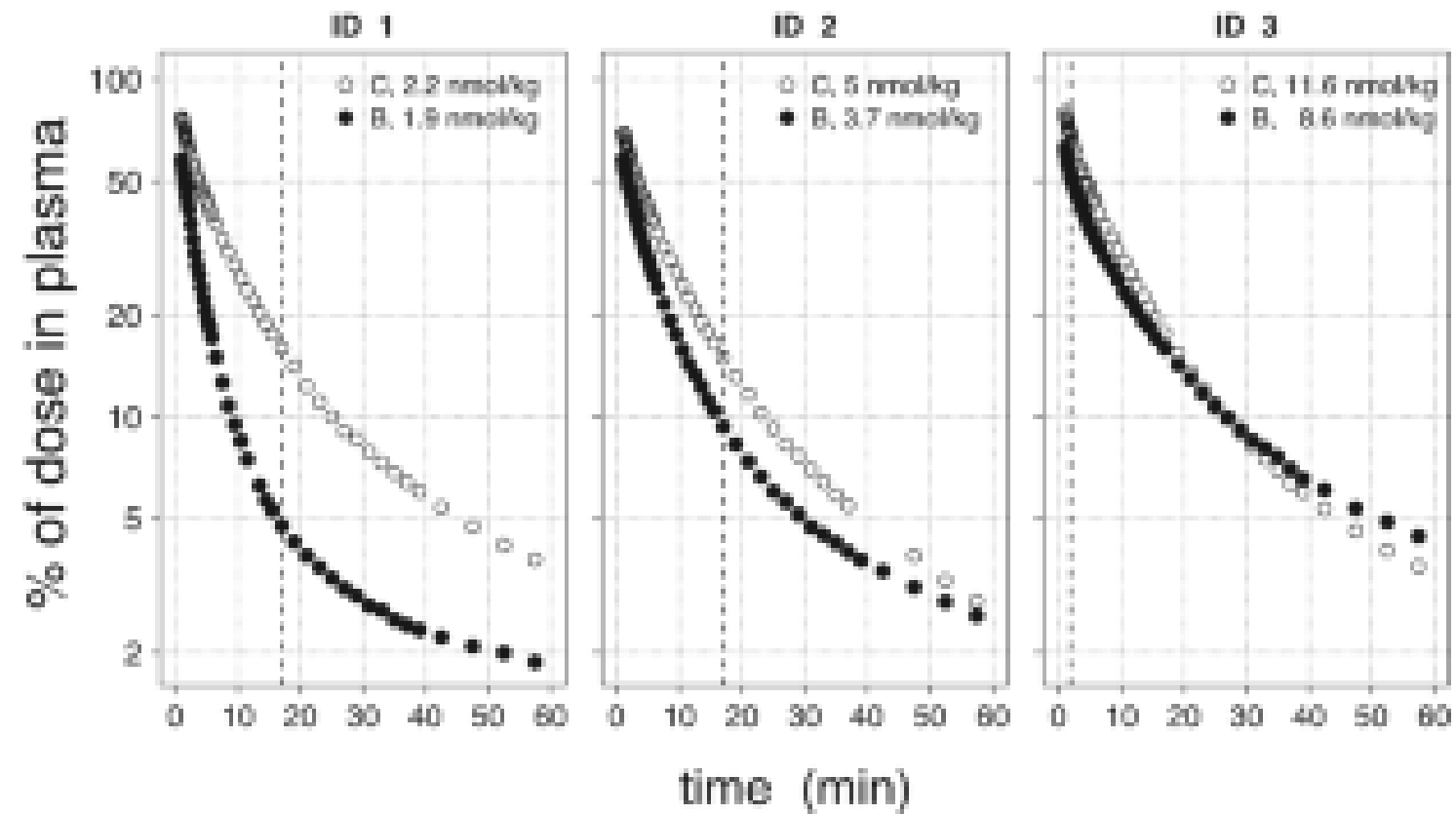


Figure 3: Figure 3

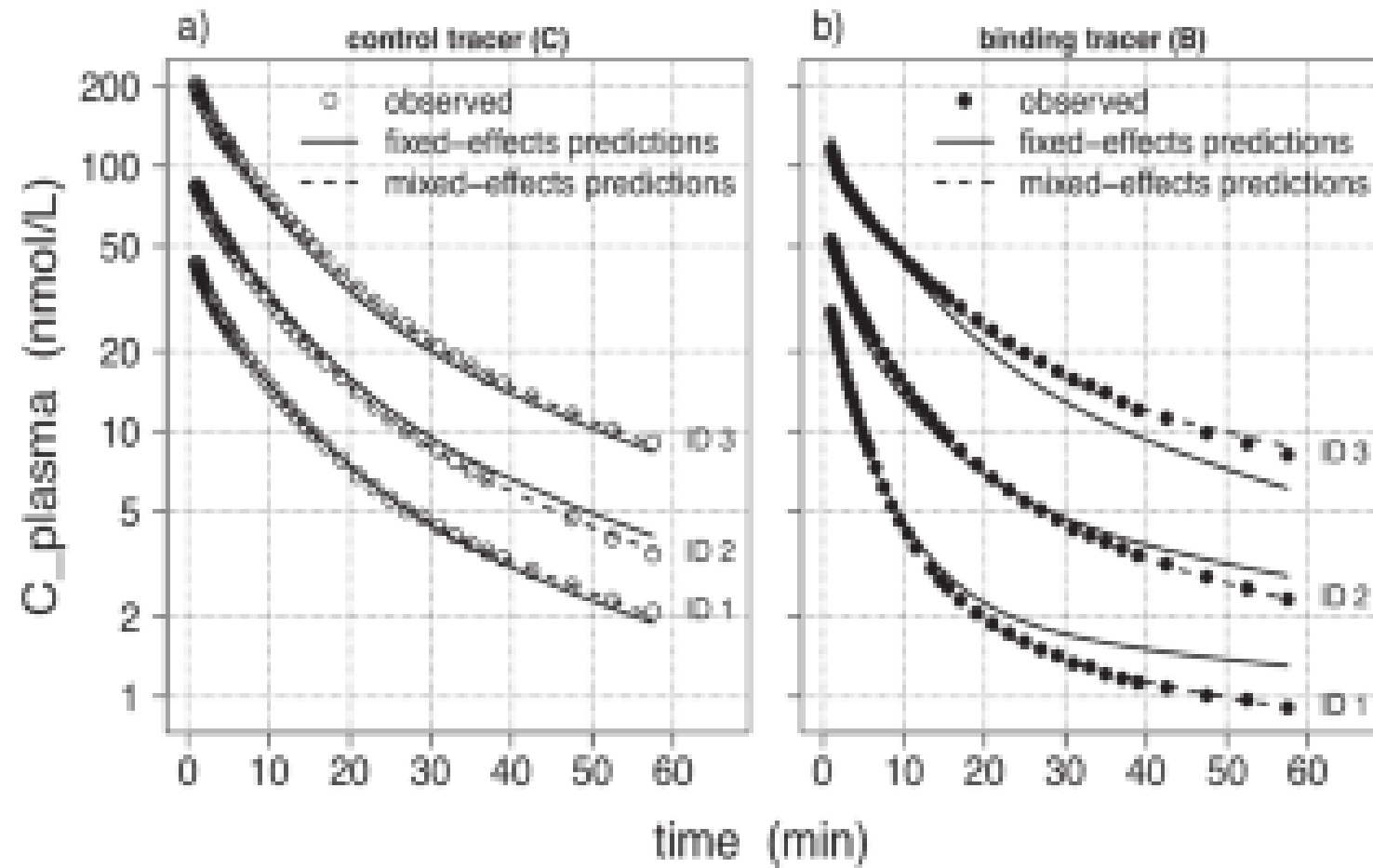


Figure 4: Figure 4

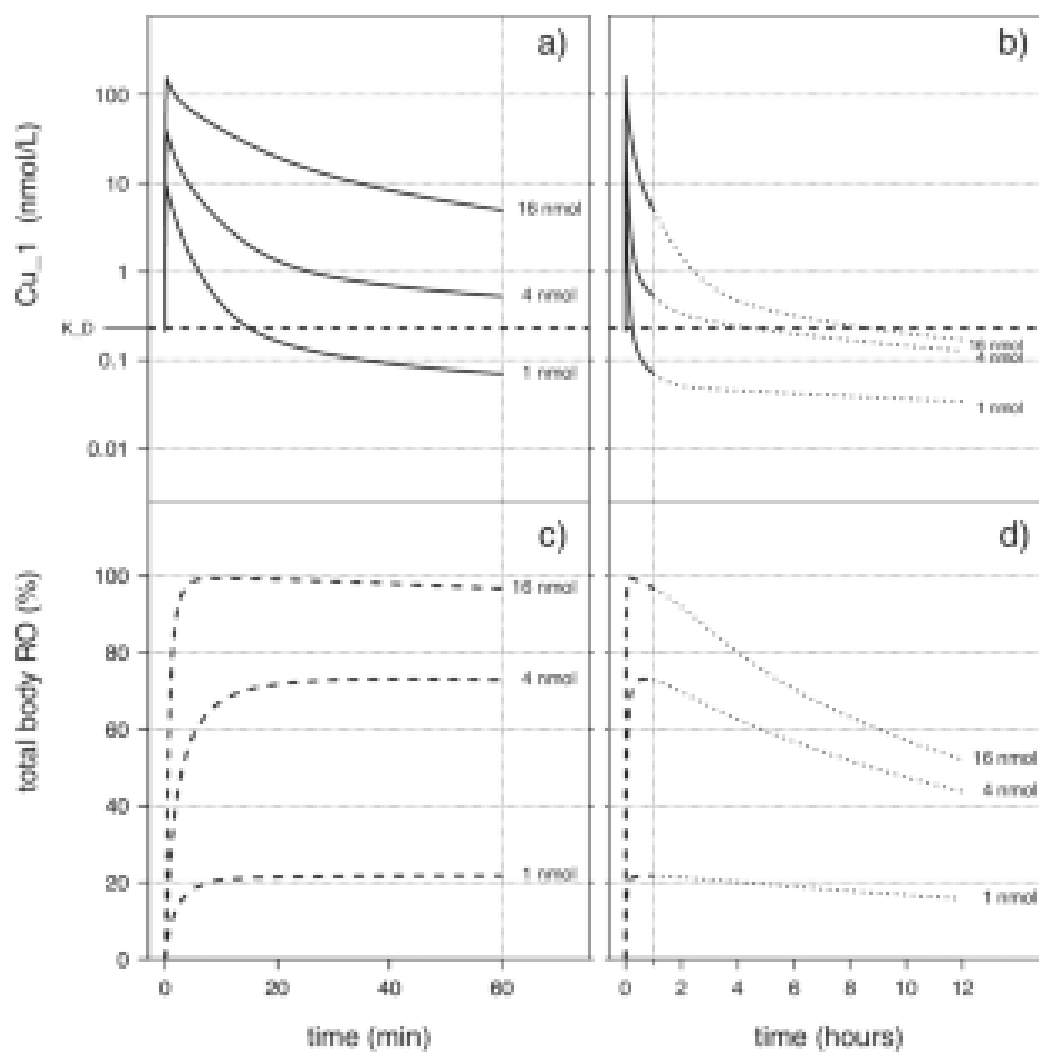


Figure 5: Figure 5